MIL-F-60541A(MU)
14 October 1966
SUPERSEDING
MIL-F-60541(MU)
14 June 1966

MILITARY SPECIFICATION

FUZE, MTSQ, M564E1 ASSEMBLY LESS BOOSTER
PELLET AND CLOSING CUP (LEAD) CHARGE

1. SCOPE

1.1 This specification covers Fuze, MTSQ, M564E1 Assembly, less booster pellet and closing cup (lead) charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-314	- Waterproofness Test for Use in Development of Fuzes
MIL-STD-350	- Jolt Test for Use in Production of Fuzes
MIL-STD-351	- Jumble Test for Use in Production of Fuzes
MIL-STD-352	- Forty-Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	- Transportation-Vibration Test for Use in Production of Fuzes
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

C8886458	- Box, Wirebound, Packing, Ammunition for Artillery Type and Rocket Fuzes (within M2A1 Box)

FSC 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

DRAWINGS

Munitions Command

F8864492	- Box, Ammunition, Metal for Artillery Type and Rocket Fuzes
C9214704	- Box, Wirebound, for Artillery Type or Rocket Fuzes
F10542753	- Fuze, MTSQ, M564E1 Assembly
IEL-10542753	- Inspection Equipment List for Fuze, MTSQ, M564E1 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications except as provided in 6.1.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.3 Dynamic regulation.- The movement subassembly shall be regulated to the frequency specified on the applicable drawing. The timing error between 2 to 100 seconds shall not exceed 0.07% while spinning at the specified revolutions per minute (rpm).

3.4 Environmental.

3.4.1 Jolt.- The fuze shall comply with MIL-STD-350.

3.4.2 Jumble.- The fuze shall comply with MIL-STD-351.

3.4.3 Forty-foot drop.- The fuze shall comply with MIL-STD-352.

3.4.4 Waterproofness.- The fuze shall comply with MIL-STD-314.

3.4.5 Transportation-Vibration.- The fuze shall comply with MIL-STD-353.

3.5 Ballistics (See 6.3.4).

3.5.1 The fuze shall not function prematurely (See 6.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.5.2 Target impact nonfunction distance.- The fuze shall not detonate the round upon impact element function when fired in the 105MM Howitzer, Charge 7 (1550 feet per second, fps) with the fuze set safe "S" on a target 220 plus 5 feet from the muzzle of the weapon.

3.5.3 Target impact function distance.- The fuze shall detonate the round upon impact when fired in the 105MM Howitzer, Charge 7 (1550 fps) with the fuze set safe "S" on a target 400 minus 5 feet from the muzzle of the weapon.

3.5.4 Ground or water impact.- The fuze shall detonate the round upon impact when fired as follows:

Weapon	Charge	Impact Angle	Setting
105MM Howitzer	1 (650 fps)	45° minimum	Safe "S"
105MM Howitzer	7 (1550 fps)	45° minimum	Safe "S"

3.5.5 Time-to-burst (ambient temperature).- The fuze shall detonate the round during flight within the time limits specified in Table I, when fired as follows:

Weapon	Charge	Setting
155MM Howitzer	1 (680 fps)	3 seconds
105MM Howitzer	7 (1550 fps)	60 seconds

3.5.6 Excess pressure.- The fuze shall detonate the round during flight when fired in the 105MM Howitzer at 112% of the normal charge 7 pressure with the fuze set at 60 seconds.

3.5.7 Low temperature.- The fuze temperature conditioned at -65° to -70° F. shall detonate the round during flight when fired as specified in 3.5.5.

3.5.8 High temperature.- The fuze temperature conditioned at +145° to +150° F. shall detonate the round during flight when fired as specified in 3.5.5.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.6.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.6.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

TABLE I

Limits for Sample Mean Time of Flight (Sec)

Weapon	Charge	Fuze Setting	Limits for Individuals (Seconds)	Sample Size 20	19	18	17	16	15
155MM How.	1	3 Sec	UL 3.48	3.253	3.255	3.257	3.259	3.261	3.263
			LL 2.88	3.105	3.103	3.101	3.099	3.097	3.095
105MM How.	7	60 Sec	UL 60.50	59.906	59.911	59.916	59.921	59.926	59.931
			LL 58.90	59.498	59.493	59.488	59.483	59.478	59.473

Maximum Limits for Sample Standard Deviation (Sec)

Weapon	Charge	Fuze Setting	Sample Size 20	19	18	17	16	15
155MM How.	1	3 Sec	.133	.134	.135	.137	.138	.139
105MM How.	7	60 Sec	.369	.372	.375	.378	.382	.386

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory, acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

340 component parts as delineated in 4.2.2
475 subassemblies as delineated in 4.2.3
151 fuze assemblies (Dwg. F10542754) as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection by the contractor and provisional acceptance at source by the Government, 125 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Three hundred and forty (340) component parts, 475 subassemblies and 26 fuze assemblies shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each component shall be tested as in 4.3.3.1 (Hardness), 4.3.3.2 (Load) and 4.3.3.4 (Bend).

4.2.2.2 Twenty (20) of each component shall be tested as in 4.3.3.3 (Load).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.7.1 (Static Load).

4.2.3.2 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.7.2 (Torque).

4.2.3.3 Twenty-five (25) of each assembly shall be tested as in 4.3.3.6 (Pull test).

4.2.4 Acceptance criteria.- Failure of any component or subassembly to comply with the requirements of the applicable drawing for the above characteristics shall be cause for rejection of the initial production sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.2.5 Fuze assemblies.- Twenty-six (26) fuze assemblies shall be subjected to the following MIL-STD- tests, with the fuzes set on safe "S".

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.4.1	4.4.19.1	4.2.5.1.1
Jumble	4	3.4.2	4.4.19.2	4.2.5.1.1
Forty-Foot Drop	5	3.4.3	4.4.19.3	4.2.5.1.2
Waterproofness	8	3.4.4	4.4.19.4	4.2.5.1.3
Transportation-Vibration	5	3.4.5	4.4.19.5	4.2.5.1.4

4.2.5.1 Acceptance criteria.- Occurrence of any of the following in any fuze shall be cause for rejection of the initial production sample.

4.2.5.1.1 Jolt and jumble

a. Any of the explosive elements burned or detonated.

b. Rotor of safety adapter armed or partially armed (See 6.2.1).

c. Firing pin in fired position.

d. Movement has run off safe position, i.e., drop-off edge of firing arm is not within scribed lines on safety disc.

e. Broken, deformed or displaced parts which render the fuze unsafe to handle or dangerous to use.

f. Detent safety mechanism of point detonating assembly not completely closed.

g. Timing disc retainer loose.

4.2.5.1.2 Forty (40) foot drop

a. Safety adapter armed or partially armed.

b. Rotor detonator burned or detonated.

4.2.5.1.3 Waterproofness.

a. Evidence of water entry

NOTE: Evidence of water entry through the safety adapter flash hole does not constitute reason for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.2.5.1.4 Transportation-Vibration.

a. Conditions listed under 4.2.5.1.1

b. Failure of any movement to meet the nonfunction and function requirements or safety adapter or detent safety mechanism to meet the nonarming and arming requirements of the applicable drawings.

4.2.6 Ballistics.- One hundred and twenty-five (125) fuzes, of the initial production sample, shall be subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Impact nonfunction distance (ambient temp.)	20	3.5.1 & 2	4.4.20.1
Impact function distance (ambient temp.)	10	3.5.1 & 3	4.4.20.2
Ground or water impact (ambient temp.)			
105MM Howitzer charge 1	5	3.5.1 & 4	4.4.20.3
105MM Howitzer charge 7	5	3.5.1 & 4	4.4.20.3
Time-to-burst (ambient temp.)			
155MM Howitzer charge 1	20	3.5.1 & 5	4.4.20.4
105MM Howitzer charge 7	20	3.5.1 & 5	4.4.20.4
Excess Pressure (ambient temp.)	25	3.5.1 & 6	4.4.20.5
Low temperature (-65 to -70° F.)			
155MM Howitzer charge 1	5	3.5.1 & 7	4.4.20.6
105MM Howitzer charge 7	5	3.5.1 & 7	4.4.20.6
High temperature (+145 to +150° F.)			
155MM Howitzer charge 1	5	3.5.1 & 8	4.4.20.6
105MM Howitzer charge 7	5	3.5.1 & 8	4.4.20.6

4.2.6.1 Acceptance criteria.- The occurrence of any of the following shall be cause for rejection of the initial production sample:

a. A premature burst in any phase (See 6.3.1)

b. Function of one or more rounds upon target impact in the Target Impact Nonfunction Distance Phase.

c. Failure of more than one round to function in the Impact Function Distance Phase.

d. Failure of more than one round to function in the Ground or Water Impact Phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.2.6.1 Cont'd

e. Any of the following in the ambient Time-to-Burst Phase at each time setting.

(1) More than one time failure (6.3.2)

(2) More than one off time (6.3.3)

(3) Failure of the sample statistics to comply with the limits specified in Table I.

f. More than two time failures (6.3.2) in the Excess Pressure Phase.

g. More than two time failures (6.3.2) in either of the Environmental Phases (Low Temperature, High Temperature).

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing examination for visual defects.

4.3.2.1 Plate No. First Lamina Drawing F10523181 covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.1 Cont'd

Defect		Method of Inspection
MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred, damaged or defective	Visual
105.	Chamfer on holes M and 20 missing or incorrect	Visual

4.3.2.2 Plate No. 2 Second Lamina Drawing F8595968, covering a detail of Drawing F10542753.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred, damaged or defective	Visual
105.	Chamfer on holes M and 20 missing or incorrect	Visual

4.3.2.3 Plate No. 2 First Lamina Drawing F8595967, covering a detail of Drawing F10542753.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Flatness incorrect	Gage
102.	Metal burred, damaged or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.4 Slide, pallet lock Drawing C10521169, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
1. Radius on extension above max.	Gage
MAJOR	
None defined	

4.3.2.5 Gear escapement Drawing D10542765 covering a detail of Drawing F10542753.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

4.3.2.6 Escapement gear and pinion assembly Drawing C10542766 covering a detail of Drawing F10542753.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Eccentricity of gear with datum	Gage
102. Perpendicularity of gear, incorrect	Gage
103. Burr	Visual

4.3.2.7 Escapement lever assembly Drawing F10542852, covering a detail of Drawing F10542753.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.7 Cont'd

Defect		Method of Inspection
MAJOR	AQL = 1.0%	
101.	Eccentricity of external radius with datum A including parallelism with datum A	Gage
102.	Metal burred, damaged or defective	Visual
103.	Surface finish improper	Visual

4.3.2.8 Arbor escapement Drawing D10535526, covering a detail of Drawing F10542753.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Perpendicularity of lever seat with datum C, incorrect	Gage
102.	Diameter of pivots, incorrect	Gage
103.	Metal burred, damaged or defective	Visual

4.3.2.9 Head Drawing F10520265, covering a detail of Drawing F10542753.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Diameter at forward datum length, incorrect	Gage
102.	Diameter at rear datum length, incorrect	Gage
103.	Eccentricity of forward diameter with datum A	Gage
104.	Eccentricity of rear diameter with datum A	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.10 Firing pin, Drawing C10521154, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of striker point, improper	Gage
102. Surface finish on angle, improper	Visual

4.3.2.11 Firing pin, Drawing C10520276, covering a detail of Drawing F10542753.

CRITICAL	
1. Length of striker exceeds max.	Gage
2. Diameter of body less than min.	Gage
3. Corner radius of body shoulder exceeds max.	Gage
MAJOR AQL = 1.0%	
101. Diameter of body exceeds max.	Gage
102. Eccentricity of striker diameter with body diameter exceeds max.	Gage
103. Diameter of striker point, improper	Gage

4.3.2.12 Firing pin assembly, Drawing C10520274, covering a detail of Drawing F10542753.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Firing pin striker damaged or deformed	Visual
102. Staking or crimping of firing pin to head inadequate	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.13 Adapter assembly, Drawing D10520269, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
1. Detonator extends above flush	Visual
MAJOR AQL = 0.40%	
101. Detonator missing or improperly assembled (red end down)	Visual
102. Crimping of detonator inadequate	Visual

4.3.2.14 Detent safety mechanism assembly, Drawing F10520250, covering a detail of Drawing F10542753.

CRITICAL	
1. Length from top of detents to base of housing less than min.	Gage
MAJOR AQL = 0.40%	
101. Improper or incomplete assembly	Manual and Visual
102. Staking of cover to housing inadequate	Visual
103. Foreign matter	Visual

4.3.2.15 Point detonating assembly, Drawing F8595988, covering a detail of Drawing F10542753.

CRITICAL	
1. Detent safety mechanism assembly missing (to be checked prior to insertion of adapter assembly)	Visual
MAJOR AQL = 0.40%	
101. Extension of adapter assembly beyond the bottom surface of head exceeds max.	Gage
102. Insecure, improper or incomplete assembly	Manual & Visual
103. Crimping of closing disc inadequate	Visual
104. Metal burred, damaged or defective	Visual
105. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.16 Cap, lower Drawing F10520255, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter incorrect at forward datum length	Gage
102. Diameter incorrect at rear datum length	Gage
103. Eccentricity of forward diameter with datum A	Gage
104. Eccentricity of rear diameter with datum A	Gage

4.3.2.17 Plug, closing Drawing F10520253, covering a detail of Drawing F10542753.

CRITICAL	
1. Small O.D. below min.	Gage
MAJOR AQL = 1.0%	
101. Location of locating pin hole incorrect	Gage

4.3.2.18 Body Drawing F10542758, covering a detail of Drawing F10542758.

CRITICAL	
1. Minor diameter of adapter cavity thread exceeds max.	Gage
MAJOR AQL = 1.0%	
101. Pitch diameter of adapter cavity thread, incorrect	Gage
102. External diameter incorrect at forward datum	Gage
103. External diameter incorrect at rear datum	Gage
104. Location of closing plug locating hole incorrect	Gage
105. Threads damaged or not full	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.19 Body Drawing F8595997, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
1. Major diameter of large external thread, improper	Gage
MAJOR AQL = 1.0%	
101. Pitch diameter of large external thread, improper	Gage
102. Major diameter of small external thread, improper	Gage
103. Pitch diameter of small external thread, improper	Gage
104. Diameter of cavity exceeds max.	Gage
105. Threads damaged or not full	Visual
106. Metal burred, damaged or defective	Visual

4.3.2.20 Firing arm assembly, Drawing C10521115, covering a detail of Drawing F10542753.

CRITICAL	
1. Fracture of upright bend of firing arm	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 1.0%	
101. Insecure weight	Manual & Visual
102. Improper or incomplete assembly	Visual
103. Metal burred, damaged or defective	Visual

4.3.2.21 Timing disc assembly, Drawing D10520268, covering a detail of Drawing F10542753.

CRITICAL	
1. Safety disc tabs not engaged with timing disc bushing slots	Visual
MAJOR AQL = 0.40%	
101. Adjusting segment radius extends beyond periphery of timing disc	Visual
102. Safety disc radius below periphery of timing disc	Visual
103. Improper or incomplete assembly	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.22 Movement subassembly (after dynamic regulation), Drawing F10523481 covering a detail of Drawing F10542753.

	Defect	Method of Inspection
CRITICAL		
1.	Firing pin fires on "not fire" position of firing arm	Gage
2.	Extension of firing arm upright above top plate less than minimum (with firing arm shaft shoulder bottomed on plate No. 2)	Gage
3.	Escapement gear not locked in properly by Lever pallet	Visual
MAJOR	AQL = 0.40%	
101.	Firing pin does not fire on "must fire" position of firing arm	Gage
102.	Firing pin protrusion less than min. in fired position	Gage
103.	Extension of firing arm upright above top plate exceeds maximum (with firing arm shaft shouldered against top plate).	Gage
104.	Insecure, improper or incomplete assembly	Manual and Visual
105.	Firing arm shaft spring protrudes	Visual
106.	Staking of body assembly screws inadequate (check prior to dynamic regulation)	Visual
107.	Arbor centerhole missing	Visual

4.3.2.23 Movement assembly - Drawing F10523480, covering a detail of Drawing F10542753.

CRITICAL		
1.	Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual and Visual
2.	Timing disc retainer insecure, damaged or missing	Manual and Visual
3.	Distance from top of top plate to top of retainer exceeds maximum (after staking of retainer)	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.23 Cont'd

Defect		Method of Inspection
MAJOR	AQL = 0.40%	
101.	Total height exceeds max.	Gage
102.	Insecure, improper or incomplete assembly	Manual and Visual
103.	Escapement lever not locked on proper side of pallet lock slides (F10523481)	Visual
104.	Escapement spring broken, deformed or unseated	Visual
105.	Foreign matter	Visual
106.	Pins or protrusions above flush of bottom surface	Visual
107.	Firing pin striker point damaged	Visual

4.3.2.24 Body, movement and lower cap assembly, Drawing F10542756 (prior to insertion of Welsh plug), covering a detail of Drawing F10542753.

CRITICAL		
1.	Drop off edge of firing arm outside of lines on safety disc (to be checked after lock ring tensioning and zero setting operations)	Visual
MAJOR	AQL = 0.40%	
101.	Dimension from locknut to top of cap less than min.	Gage
102.	Eccentricity of arbor to datum "A"	Gage
* 103.	Setting pin extension below timing disc lug less than minimum	Gage
104.	Insecure, improper or incomplete assembly	Manual and Visual
* 105.	Setting pin not engaged in timing disc lug slot	Visual
106.	Body assembly screw(s) inadequately staked	Visual

* To be checked prior to assembly of lower cap

4.3.2.25 Detonator screw assembly, drawing C10520294, covering a detail of Drawing F10542753.

CRITICAL		
1.	Detonator extends above flush of detonator screw	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.25 Cont'd

Defect		Method of Inspection
MAJOR	AQL = 1.0%	
101.	Overall length exceeds maximum	Gage
102.	Insecure, improper or incomplete assembly	Manual and Visual
103.	Green end of detonator not at bottom (thread end)	Visual
104.	Crimping inadequate	Visual

4.3.2.26 Fuze subassembly, less safety adapter, prior to assembly of closing plug assembly, Drawing F10542755 covering a detail of Drawing F10542753.

CRITICAL		
1.	Escapement lever pallet(s) missing, damaged or deformed	Visual
MAJOR	AQL = 0.40%	
101.	Detonator screw above flush of body surface	Visual
102.	Detonator screw assembly inserted improperly	Visual
103.	Staking of detonator screw assembly inadequate	Visual

4.3.2.27 Fuze subassembly, less safety adapter, Drawing F10542755, covering a detail of Drawing F10542753, with closing plug assembly inserted.

CRITICAL		
None defined		
MAJOR	AQL = 0.40%	
101.	Distance from flange to top exceeds max.	Gage
102.	Insecure, improper or incomplete assembly	Manual and Visual
103.	Relay improperly inserted in closing plug (closed end not visible)	Visual
104.	Relay extends beyond flush	Visual
105.	Spinning or crimping of relay inadequate	Visual
106.	Staking of assembly screws inadequate	Visual
107.	Metal burred, damaged or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.28 Rotor assembly, Drawing D8594658 covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Insecure assembly	Manual
102. Detonator installed incorrectly (with yellow end at crimp)	Visual
103. Detonator extends above flush of rotor surface	Visual
104. Rotor gear pin extends above flush of rotor surface	Visual
105. Staking or crimping of detonator assembly inadequate	Visual
106. Metal burred, damaged or defective	Visual

4.3.2.29 Safety adapter assembly, Drawing F10534290, covering a detail of Drawing F10542753.

CRITICAL	
1. Interference condition which precludes return of either spring-loaded detent from the fully open position to the position of engagement in the rotor recess (check prior to nonarming test)	Visual and Manual
2. Rotor not locked in the unarmed position (see 6.2.2) (to be checked immediately prior to assembly into fuze)	Visual and Manual
MAJOR AQL = 0.40%	
101. Length from base to movement plate exceeds max.	Gage
102. Protrusion of screws, rotor shaft and pins above plate exceeds max.	Gage
103. Staking of plate screw(s) inadequate	Visual
104. Metal burred, damaged or defective	Visual
105. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.30 Fuze, subassembly, Drawing F10542754, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
1. Fuze not set on Safe "S" (to be checked immediately prior to packaging)	Visual
2. Safety adapter rotor armed or partially armed as viewed through the flash hole (to be checked immediately prior to packaging)	Visual
MAJOR AQL = 0.40%	
101. Total length exceeds max.	Gage
102. Length from base to flange exceeds max.	Gage
103. Major diameter of fuze body external thread, improper	Gage
104. Pitch diameter of fuze body external thread, improper	Gage
105. Major diameter of safety adapter external thread, improper	Gage
106. Pitch diameter of safety adapter external thread, improper	Gage
107. Insecure, improper or incomplete assembly	Manual and Visual
108. Threads damaged or defective	Visual
109. Marking missing, incorrect or illegible	Visual
110. Dust cover (over flash-hole) missing	Visual

4.3.2.31 Closing cup, Drawing B8595508, covering a detail of Drawing F10542753.

CRITICAL	
1. Split, cut or crack	Visual
MAJOR AQL = 1.0%	
101. Thickness of base, improper	Gage
102. Length under flange, improper	Gage
103. Outside diameter exceeds max.	Gage
104. Score, wrinkle or dent	Visual
105. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.32 Booster cup, Drawing B8595509, covering a detail of Drawing F10542753.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Inside depth, improper	Gage
102.	Pitch diameter of thread, improper	Gage
103.	Minor diameter of thread exceeds max.	Gage
104.	Outside diameter, improper	Gage
105.	Effective length of thread, less than min.	Gage
106.	Metal defective	Visual
107.	Cut, score or gouge on inside wall	Visual
108.	Foreign matter	Visual

4.3.2.33 Box, Ammunition, Metal Drawing F8864492.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Moisture inside box (check prior to packing)	Visual
102.	Gasket damaged or unseated (check prior to packing)	Visual
103.	Polystyrene supports/fuzes packed inadequately	Visual and Manual
104.	Dessicant bags/containers damaged, loose or missing	Visual and Manual

MINOR AQL = 2.5%

201.	Box damaged or deformed	Visual
202.	Marking missing, incorrect or illegible	Visual

4.3.2.34 Box, Wirebound, Packing, Ammunition for Artillery Type and Rocket Fuzes (within M2A1 Box) Drawing C8886458.

CRITICAL

None defined

MAJOR

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.34 Cont'd

Defect		Method of Inspection
MINOR	AQL = 2.5%	
201.	Contents loose	Visual
202.	Boxes improperly arranged	Visual
203.	Sleeve or spacer missing	Visual
204.	Strapping missing or loose	Visual
205.	Marking missing, incorrect or illegible	Visual

4.3.2.35 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of the firing arm upright is not permitted at any time.

a. Laminated Plates No. 1 and No. 2 Assembly, Drawing F10542853.

(1) Application of lubricant

b. Safety Adapter Assembly, Drawing F10534290

(1) Application of lubricant

c. Movement subassembly, Drawing F10523481

(1) Application of lubricant

(2) Top and bottom assembly screws staked prior to dynamic regulation

(3) Mainspring wound before assembly of screw

(4) Dynamic regulation operation

(5) Movement fully wound after dynamic regulation

d. Escapement lever assembly, Drawing F10542852 securing escapement lever spring

e. Movement assembly, Drawing F10523480

(1) Locknut torqued prior to timing disc torque test

f. Body, movement and lower cap assembly, Drawing F10542756

(1) Application of lubricant
(2) Insertion of "O" ring between lower cap and body
(3) Lockring torque operation
(4) Drilling operation for lockring setscrew
(5) Zero setting procedure
(6) Application of sealant

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.35 (Cont'd)

g. Fuze subassembly, less safety adapter, Drawing F10542755

(1) Application of lubricant
(2) Application of sealant
(3) Insertion of dessicant pellet and cushion
(4) Insertion of "O" ring between head and lower cap
(5) Head torque operation
(6) Fuze setting torque

h. Fuze, subassembly, Drawing F10542754.

(1) Application of sealant to safety adapter threads
(2) Safety adapter tightening operation

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions 4.4.

4.3.3.1 Hardness (Destructive).- Twenty samples from each lot of the following components shall be subjected to the hardness test. Where plating is required, the sample parts tested shall be unplated, where feasible. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of components, as applicable (see 4.4.1).

Component	Drawing Number
Arbor, escapement	D10535526
Gear, escapement	D10542765
Pinion, escapement	D10521101
Pinion, No. 1	D10521119
Pinion, No. 2	D10521120
Pinion, No. 3	D10521118
Pinion, No. 4	C10521123
Wheel, escape (Safety Adapter Assy)	C8595539
Arm, firing (see 4.3.3.4)	D10521116
Guide, Adjusting	C8594273
Lever, escapement	F10520340
Spring, hammer	C10521135
Spring, timing disc (see 4.3.3.2)	C10521161
Washer, Bearing	B10520292
Washer, Timing Disc	B8594267
Washer, Torque	C10520291

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.2 Load - Timing disc spring (destructive).- The same 20 timing disc springs subjected to the hardness test shall be subjected to the load specified on Drawing C10521161. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of timing disc springs (see 4.4.2).

4.3.3.3 Load - Support cup (destructive).- Twenty support cups from each lot shall be subjected to the load specified on Drawing C10521142. Failure of any support cup to comply with the requirement shall be cause for rejection of the lot of support cups (see 4.4.3).

4.3.3.4 Bend - Firing arms (destructive).- The same 20 firing arms that comply with the hardness requirement shall be subjected to the bend specified on Drawing D10521116. Failure of any firing arm to comply with the requirement shall be cause for rejection of the lot of firing arms (see 4.4.4).

4.3.3.5 Compression - Springs (destructive).- Twenty firing pin springs from each lot shall be subjected to the compression specified on Drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of firing pin springs (see 4.4.5).

4.3.3.6 Pull test - Escapement spring (nondestructive).- Twenty-five (25) escapement springs from each lot of escapement lever assemblies shall be subjected to the pull test specified on Drawing F10542852. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of escapement lever assemblies (see 4.4.6).

4.3.3.7 Static load and Torque (destructive).

4.3.3.7.1 Static load.- Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.7).

Static Load	Drawing Number
Disc and Stud Assembly	D8593103
Escapement Gear and Pinion Assembly	C10542766
Escapement Lever Assembly	F10542852
Firing Arm Assembly (Test Pin and Arm)	C10521115
No. 1 Gear and Pinion Assembly	D10521102
No. 2 Gear and Pinion Assembly	C10521104
No. 3 Gear and Pinion Assembly	C10521106
No. 4 Gear and Pinion Assembly	C10521111
No. 5 Gear and Arbor Assembly	C10522355
Detent Safety Mechanism Assembly	F10520250

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

Static Load	Drawing Number
Pallet Lock Assembly	B10521168
Lever Assembly (pin and spring)	B10521148
Escape Wheel Assembly (Safety Adapter Assembly)	C8595538
No. 1 Gear and Pinion Assembly (Safety Adapter Assembly)	C8594673
No. 2 Gear and Pinion Assembly (Safety Adapter Assembly)	C8594681
Balance Staff Assembly	D8594614

4.3.3.7.2 Torque.- Twenty-five samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot subassemblies, as applicable (see 4.4.8).

Torque	Drawing Number
Escapement Lever Assembly	F10542852
No. 5 Gear and Arbor Assembly	C10522355

4.3.3.8 Setting pin load (nondestructive).- The setting pin of fifty housing and hammer spring assemblies from each lot shall be subjected to the load specified on Drawing C10520266. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of housings (see 4.4.9).

4.3.3.9 Hammerspring deflection (nondestructive).- The hammerspring of each housing and hammerspring assembly shall be subjected to the load specified on Drawing C10520266. Failure to comply with the requirements shall be cause for rejection of the assembly (see 4.4.10).

4.3.3.10 Kinetic energy of firing pin (nondestructive).- Fifty movement subassemblies from each lot shall be tested to determine compliance with the requirement specified on Drawing F10523481. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of movement subassemblies (see 4.4.12).

4.3.3.11 Spin function - (movement subassembly) (nondestructive).- The pallet lock and safety lever assemblies within each movement subassembly shall be tested to determine compliance with the function requirement of Drawing F10523481. Failure to comply with the requirement shall be cause for rejection of the subassembly (see 4.4.13).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.12 Spin nonfunction (movement subassembly) (nondestructive).- The movement subassembly shall be tested to determine compliance with the nonfunction requirement of Drawing F10523481. Failure of the lever lock slides to remain in the locked position shall be cause for rejection of the subassembly (see 4.4.14).

4.3.3.13 Timing disc torque (nondestructive).- The timing disc of each movement assembly shall be subjected to the torque specified on Drawing F10523480. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.15).

4.3.3.14 Arming (safety adapter assembly) (nondestructive).- Each safety adapter shall be tested to determine compliance with the arming requirement of Drawing F10534290. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.16).

4.3.3.15 Nonarming (safety adapter assembly) (nondestructive).- Each safety adapter shall be tested to determine compliance with the nonarming requirement of Drawing F10534290. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.17).

4.3.3.16 Arming (detent safety mechanism) (nondestructive).- Each detent safety mechanism shall be tested to determine compliance with the arming requirement of Drawing F10520250. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.18).

4.3.3.17 Nonarming (detent safety mechanism assembly) (nondestructive).- Each detent safety mechanism shall be tested to determine compliance with the nonarming requirement of Drawing F10520250. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.18).

4.3.3.18 Environmental tests (lots) (destructive).- Twenty-six (26) fuze assemblies from each production lot shall be subjected to the following MIL-STD- tests, with the fuzes set safe "S".

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.4.1	4.4.19.1	4.2.5.1.1
Jumble	4	3.4.2	4.4.19.2	4.2.5.1.1
Forty-foot drop	5	3.4.3	4.4.19.3	4.2.5.1.2
Waterproofness	8	3.4.4	4.4.19.4	4.2.5.1.3
Transportation-Vibration	5	3.4.5	4.4.19.5	4.2.5.1.4

4.3.3.18.1 Acceptance criteria.- Lot acceptance criteria shall be as specified in 4.2.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.19 Ballistics (destructive).- Eighty (80) fuzes from each lot shall be subjected to the following ballistic test phases:

Phases (ambient temperature)	Quantity	Requirement	Test Method
Impact Nonfunction Distance	20	3.5.1 & 2	4.4.20.1
Impact Function Distance	10	3.5.1 & 3	4.4.20.2
Ground or Water Impact			
105MM Howitzer charge 1	5	3.5.1 & 4	4.4.20.3
105MM Howitzer charge 7	5	3.5.1 & 4	4.4.20.3
Time-to-Burst			
155MM Howitzer charge 1	20	3.5.1 & 5	4.4.20.4
105MM Howitzer charge 7	20	3.5.1 & 5	4.4.20.4

4.3.3.19.1 Acceptance criteria.- The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

(a) A premature burst in any phase (see 6.3.1)

(b) Function of one or more rounds upon target impact in the Impact Nonfunction Distance Phase

(c) Failure of more than one round to function in the Impact Function Distance Phase

(d) Failure of more than one round to function in the Ground or Water Impact Phase

(e) Any of the following in the Ambient Time-to-Burst Phase at any time setting

(1) More than one time failure (6.3.2)

(2) More than one off time (6.3.3)

(3) Failure of the sample statistics to comply with the limits specified in Table I.

4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-10542753. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.1 Hardness.- Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual value.

4.4.2 Load (timing disc spring).- The test shall be performed using the values as specified on Drawing C10521161. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.3 Load (support cup).- The test shall be performed using the values specified on Drawing C10521142. Support cups which have been subjected to the applied load test shall be scrapped.

4.4.4 Bend (firing arm).- The test shall be performed as specified on Drawing D10521116. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.5 Compression.- The firing pin springs shall be compressed for the length of time specified on Drawing C10521132. The springs that have been subjected to the compression test shall be scrapped.

4.4.6 Pull test (escapement spring).- The test shall be performed using the values as specified on Drawing F10542852.

4.4.7 Static load.- Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

4.4.8 Torque.- Escapement lever assemblies shall be tested prior to insertion of the escapement spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

4.4.9 Setting pin load.- The setting pin within the housing and hammerspring assemblies shall be tested for the minimum load requirement of Drawing C10520266. The dimension from the bottom of the pin to surface "C" shall be checked before and after load application.

4.4.10 Hammerspring deflection (housing and hammerspring assembly).- The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on Drawing C10520266. The hammerspring must return to surface "C" when released.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.11 Dynamic regulation monitoring.- The following procedure shall be observed. The movement subassemblies shall be regulated, using the operational procedures furnished with the equipment. The movement subassemblies shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6,000 ± 15 rpm. Timekeeping and performance shall be checked over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run. Inspection shall be performed on a monitoring basis only (paragraph 4.3.2.35).

4.4.12 Kinetic energy of firing pin (movement subassembly).-The assembly shall be placed in a fixture and test for kinetic energy by releasing the firing pin from the cocked position.

4.4.13 Spin function (movement subassembly).- The spin function test shall be performed on a completely wound subassemblies. The subassemblies shall be rotated at the rpm specified on Drawing F10523481 in a clockwise direction when viewed from the bottom, for a period of not less than 10 seconds at constant rpm. Compliance with the requirement shall be determined after the test by the position of the safety lever (contacting the periphery of the collar). At the conclusion of the test, each mechanism shall be rewound. The function test shall be performed prior to the nonfunction test.

4.4.14 Spin nonfunction (movement subassembly). The spin nonfunction test shall be performed on completely wound subassemblies. The subassemblies shall be rotated at the rpm specified on Drawing F10523481 in a clockwise direction when viewed from the bottom for a period of not less than 3 seconds at constant rpm to determine compliance with the requirement (neither lever lock slide releases the escapement lever). This test shall be the last spin test to which the movement is subjected.

4.4.15 Timing disc torque.- The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance with the requirements of Drawing F10523480.

4.4.16 Arming (safety adapter assembly).- The unarmed safety adapter shall be rotated in a counterclockwise direction when viewed from the movement plate end, at the rpm specified on Drawing F10534290. The arming test shall be performed prior to the nonarming test. At the conclusion of the test, the rotor shall be returned to the unarmed position (see 6.2.1 and 6.2.2).

4.4.17 Nonarming (safety adapter assembly).- The unarmed safety adapter (see 6.2.2) shall be rotated in a counterclockwise direction when viewed from the movement plate end, at the rpm specified on Drawing F10534290. The nonarming test shall be the last spin test to which the safety adapter is subjected prior to assembly into the fuze.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.18 Arming and nonarming (detent safety mechanism).- The assembly shall be rotated in the counterclockwise direction when viewed from the cover plate and at the rpm specified on Drawing F10520250. The arming test shall be performed prior to the nonarming test.

4.4.19 Environmental test procedures.- The component parts of the fuze, which have been disassembled and inspected following tests, shall be scrapped. Where spin testing is required, the nonarming test shall be performed prior to the arming tests (Drawing F10542754).

4.4.19.1 Jolt.- The test shall be performed in accordance with the procedure specified in MIL-STD-350. The fuze shall be disassembled and inspected as follows:

(a) Remove dust seal and check position of adapter rotor.

(b) Remove head assembly and check movement safety disc relative to the firing arm.

(c) Remove safety adapter. Check detents and position of rotor.

(d) Remove closing plug, check relay.

(e) Remove lower cap and housing. Check timing disc retainer.

(f) Remove movement assembly from body. Check position of firing pin, pallet lock slides, check M47 Detonator.

(g) Remove P.D. adapter assembly and safety detent mechanism. Check position of detents, check M50 Detonator.

(h) Disassemble subassemblies to the extent that all critical parts can be examined, i.e., all gears, pinions and springs, etc., for damage.

4.4.19.2 Jumble.- The test shall be performed in accordance with the procedure specified in MIL-STD-351. The fuze shall be disassembled and inspected as outlined in paragraph 4.4.19.1.

4.4.19.3 Forty-foot drop.- The test shall be performed in accordance with the procedure specified in MIL-STD-352. The fuze shall be disassembled and inspected as follows:

(a) Remove dust seal and check position of adapter rotor.

(b) Remove safety adapter and move rotor to armed position, check Detonator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.19.4 Waterproofness.- The test shall be performed in accordance with the procedure specified in MIL-STD-314. The safety adapter flash hole shall be adequately sealed for the test. The fuze shall be disassembled and inspected as follows:

(a) Remove safety adapter, head, lower cap, check all visible internal surfaces for evidence of water entry.

4.4.19.5 Transportation-vibration.- The test shall be performed in accordance with the procedure specified in MIL-STD-353, as follows:

(a) Parallel to longitudinal axis, nose up

(b) First transverse axis tube parallel to a line running through the 100 second marking.

(1) Remove dust seal and check position of adapter rotor

(2) Remove head assembly and check movement safety disc relative to the firing arm.

(3) Remove safety adapter. Spin for nonarming and arming requirements.

(4) Remove closing plug, check relay

(5) Remove lower cap and housing. Check timing disc retainer

(6) Remove movement assembly from the body. Check position of firing pin, and M47 Detonator. Spin for nonfunction and function requirements.

(7) Remove P.D. adapter assembly and detent safety mechanism. Spin detent safety mechanism for nonarming and arming requirements.

(8) Disassemble subassemblies to the extent that all critical parts can be examined, i.e., all gears, pinions and springs, etc. for damage.

4.4.20 Ballistics.- The complete fuze assembly, Drawing F10542753 shall be assembled to a suitable projectile and fired as follows (see 6.3.4).

4.4.20.1 Impact nonfunction distance phase.- The sample fuzes shall be fired as specified in 3.5.2 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.19 as appropriate. Observation of impact element function is required.

4.4.20.2 Impact function distance phase.- The sample fuzes shall be fired as specified in 3.5.3 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.19 as appropriate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.20.3 Ground or water impact phase. - The sample fuzes shall be fired as specified in 3.5.4 under the conditions as delineated in 4.2.6 or 4.3.3.19 as appropriate.

4.4.20.4 Time-to-burst phase. - The sample fuzes shall be fired as specified in 3.5.5 under the conditions delineated in 4.2.6 or 4.3.3.19 as appropriate. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The average of the two values shall be rounded off to the second decimal place using the method prescribed by the American Society for Testing Materials.

4.4.20.4.1 The sample mean and standard deviation values shall be calculated by:

$$\overline{X} = \frac{\Sigma X}{n}$$

$$\sigma = \sqrt{\frac{n\Sigma X^2 - (\Sigma X)^2}{n\,(n-1)}} \quad \text{or equivalent}$$

Where:

X is the individual values

n is the number of values

All values of the sample shall be used to calculate the sample mean and standard deviation. The standard deviation value obtained shall then be used in the determination of outliers (non-normal values) by using the criteria $\overline{X}$ plus and minus four standard deviations. A value judged to be an outlier shall be excluded and the remaining values in the sample used to recalculate $\overline{X}$ and σ.

4.4.20.5 Excess pressure phase. - The sample fuzes shall be fired as specified in 3.5.6 under the conditions delineated in 4.2.6. Time-to-burst shall be measured in the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.20.6 Extreme temperature phase. - The sample fuzes shall be fired as specified in 3.5.5 under the conditions delineated in 4.2.6. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.20.6.1 Temperature conditioning (-65° F. and +145° F.).- The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus 65°F or plus 145°F. as applicable, until stabilization of the fuze temperature is reached (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

After stabilization, the fuze shall be saturated at minus 65° F or plus 145° F as applicable, for one hour, minimum.

5. PREPARATION OF DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level "B" or level "C".

5.1.1 Level "B". - The packaging shall be in accordance with Drawing F8864492, except air pressure test is not required.

5.1.2 Level "C". - Packing shall be in accordance with Drawing C9214705.

5.2 Packing

5.2.1 Level "B". - Packing shall be in accordance with Drawing C8886458.

5.2.2 Level "C". - Packing shall be in accordance with Drawing C9214704.

5.3 Marking. - Marking shall be as required by the applicable packaging and packing drawings.

6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 and those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/unarmed safety adapter

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

6.2.1 Armed. - Rotor is locked by the spring loaded lock pin in the position in which the detonator is fully visible through the flash hole.

6.2.2 Unarmed. - Rotor locked by both spring loaded detents with the detonator not at all visible through the flash hole.

6.3 Ballistic parameters

6.3.1 Premature burst. - Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun.

b. The round detonates before 45 seconds when set at 60 seconds or before 1.9 seconds when set at 3 seconds.

c. The round detonates before impact when set on safe "S" in the Impact Nonfunction Distance Phase, Impact Function Distance Phase, or the Ground or Water Impact Phase.

6.3.2 Time failure. - A time failure is considered to have occurred if the round fails to detonate during flight.

6.3.3 Off-time. - An off-time is an individual round burst time not within the limits for individual rounds specified in Table I.

6.3.4 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated fuze may be used for this purpose.

6.5 Data cards. - Data card information shall be as specified in MIL-STD-1167.

NOTE: Packaging and packing are reusable.

1. Level "C" packaging and packing is intended primarily where fuzes are shipped for additional loading and then have to be shipped again. The cost of this packaging is considered economical enough to eliminate returns to the original metal parts producer.

2. Level "B" packaging and packing is intended for indeterminate storage and it may be used when fuzes are shipped to a loading plant for final assembly of the booster and fuzes will then be packed for troop issue as a separate item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

6.6 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging required (see 5.1).

c. Level of packing required (see 5.2).

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390-A460

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.